File permissions in Linux

Project description

In this activity, I worked within the /home/researcher2/projects directory to check, interpret, and modify file and directory permissions using Linux commands. I examined existing permissions, analyzed 10-character permission strings, and applied chmod commands to meet organizational authorization requirements.

Check file and directory details

To check the permissions of all files, including hidden files, I used the following Linux command:

ls -la

This command displayed a detailed list of every file and subdirectory in the projects directory. Based on the provided permissions document, the directory contains five regular files, one hidden file (.project_x.txt), and a subdirectory named drafts.

Describe the permissions string

Each file or directory in Linux displays a 10-character permissions string. For example, the file project_t.txt has the permissions string:

-rw-rw-r--

The first character (-) indicates it is a regular file. • Characters 2–4 (rw-) show the user has read and write access. • Characters 5–7 (rw-) show the group has read and write access. • Characters 8–10 (r--) show all others have read access only.

Change file permissions

The organization does not allow write permissions for ‘other’ on any file. From the permissions list, project_k.txt incorrectly had write access for ‘other’. I corrected this using the command:

chmod o-w project_k.txt

After running ls -la again, the write permission was removed from the ‘other’ category.

Change file permissions on a hidden file

The file .project_x.txt is a hidden file because it begins with a period. It should not have write permissions for anyone, but the user and group should still be able to read it. I updated its permissions with the following commands:

chmod u-w .project_x.txt chmod g-w .project_x.txt chmod g+r .project_x.txt

These commands removed write permissions and ensured proper read access for user and group.

Change directory permissions

Only the researcher2 user should be allowed to access the drafts directory. To ensure this, I removed execute permissions from the group and others using:

chmod go-x drafts

This prevented anyone besides researcher2 from entering the directory.

Summary

In this activity, I inspected and updated file permissions to match organizational requirements. I used ls -la to review existing permissions and applied chmod to modify access rights for files, hidden files, and directories. This ensured that sensitive project files and folders were secured according to authorization needs.